Yuhang Gu (Thomas Gu)

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LinkedIn | Google Scholar | GitHub
Ph.D. Candidate, McMaster University | Machine Learning · Optimization · Data

PROFILE OVERVIEW

Ph.D. researcher applying **machine learning**, **deep learning**, and **optimization** to large, noisy sensor and time-series data, with strong **Python/MATLAB** programming, model development and **evaluation**, and exact/heuristic algorithms validated in simulation. Built quantified, benchmark-beating models for **AI**-driven prediction and decision-making in intelligent transportation systems; seeking **software, AI, and data** roles.

Machine learning & AI	Classification, anomaly detection, deep learning (MLP, CNN-LSTM-Attention), neuro-fuzzy inference (ANFIS), statistical/predictive modeling.
Optimization & algorithms	Exact/combinatorial optimization, search-space pruning, rolling-horizon control, metaheuristics (GA, PSO, Tabu Search).
Data & systems	Sensor/time-series data processing, feature engineering, simulation, model evaluation, data-driven decision-support.

EDUCATION

McMaster University, Hamilton, Canada – Ph.D. Candidate, Civil Engineering *2022 – 2026 (expected)*
Focus: **machine learning**, **optimization**, data analytics, and simulation. **Coursework GPA: 4.0/4.0**. Expected graduation *August 2026*; available *September 2026*.

Hohai University, Nanjing, China – M.Phil., Transportation Planning and Management *2019–2022*
GPA: 89.7/100; Rank: 1/15. Thesis: multimodal logistics route optimization.

Guilin University of Electronic Technology, Guilin, China – B.Eng., Traffic Engineering *2015–2019*
GPA: 85.8/100; Rank: 2/79.

RESEARCH EXPERIENCE

Graduate Researcher, McMaster University, Hamilton, Canada *2022–Present*

- **Machine learning:** engineered a supervised classification pipeline in **scikit-learn** (feature extraction, model selection, cross-validated evaluation) on noisy sensor data; the random-forest model reached *0.83 accuracy and 0.825 AUC*, best of four algorithms benchmarked.
- **Deep learning:** built and benchmarked deep-learning models in **TensorFlow** (MLP vs. CNN-LSTM-Attention); integrated the best-performing MLP with neuro-fuzzy inference (ANFIS) for queue-length estimation, achieving *1.65–9.25% MAPE* and outperforming conventional baselines.
- **Optimization & algorithms:** designed an exact combinatorial optimizer that uses learned constraints to prune the search space *by 62%*; on a three-intersection corridor network it reduced average delay *by 72%* versus a throughput-optimal baseline while matching its throughput within *3.1%*.
- **Simulation & data processing:** developed and calibrated microscopic simulations (VISSIM, SUMO) across many scenarios to stress-test models under noise, missing data, and edge cases.

Researcher, Hohai University, Nanjing, China *2019–2022*

- **Optimization & modeling:** developed metaheuristic routing, facility-location, and network-design models (VRP/VRPTW; genetic algorithms, particle swarm optimization) for logistics and transit systems.

SELECTED DATA SCIENCE PROJECTS

Applied **machine learning**, **optimization**, and **data-analytics** engineering to real sensor, camera, and operational data.

Real-Time Queue Detection – CUBIC, RTA 1.3 *2022–2023*
Contributed to an analytics **workflow** on **camera/detector** data (GridSmart) for long-queue detection, supporting central control, delay **prediction**, and signal-timing decisions.

Large-Scale Data Analytics – Guilin University of Electronic Technology *2017–2019*
Data analytics: large-scale taxi/floating-car origin–destination (travel-demand) analysis on a **MariaDB** database system; demand-pattern and trend characterization. *(Undergraduate research; software copyright 2019SR0559116.)*

TECHNICAL SKILLS

Programming & data	Python, MATLAB, Git, SPSS, ECharts.
Frameworks & libraries	TensorFlow/Keras, scikit-learn, PyTorch, pandas, NumPy, Matplotlib/Seaborn.
Machine learning & AI	Random forest, SVM, Gaussian mixture models, MLP, CNN-LSTM-Attention, ANFIS, anomaly detection.
Optimization & algorithms	Exact/combinatorial optimization, search-space pruning, genetic algorithms, particle swarm optimization, Tabu Search.
Simulation & modeling	Microscopic simulation (VISSIM, SUMO), feature engineering, model evaluation, GIS (ArcGIS, TransCAD).
Domain background	Intelligent transportation systems, logistics and routing (VRP/VRPTW).

PROFESSIONAL EXPERIENCE

Teaching Assistant, McMaster University, Hamilton, Canada *Fall 2023; Fall 2025*

Teaching & mentoring (CIVENG 2E03, programming): led **MATLAB** computation tutorials, graded and mentored students, and translated complex technical concepts into clear instructional materials.

Teaching Assistant, McMaster University, Hamilton, Canada *Winter 2023; Fall 2024*

Project coordination & design review (CIVENG 4X06A/B, capstone): guided capstone teams through scoping, design review, technical feedback, and professional communication.

- **Technical review:** peer reviewer for *Transportation Research Part A*, *Transportmetrica A*, *IEEE Access*, *Cluster Computing*, *Scientific Reports*, *Measurement and Control*, etc., evaluating methodology, rigor, and reproducibility.

SCHOLARLY PUBLICATIONS AND CONTRIBUTIONS

Thomas Gu (Yuhang Gu) – Google Scholar (June 2026): 161 citations, h-index 5, i10-index 3.

First-Author Journal Articles

1. **Gu, Y.**; Razavi, S.; Yang, H. Headway Anomaly Detection for Lane-Based Queuing Status Classification Using Detector Presence Data: Comparative Study. *Transportation Research Record*, 2025. doi:10.1177/03611981251332259.
2. **Gu, Y.**; Razavi, S.; Habibi, S. Cycle Maximum Queue Length Estimation: An Integrated Deep Learning and Adaptive Neuro-Fuzzy Inference System Framework. *IEEE Access*, 2024. doi:10.1109/ACCESS.2024.3493753.

Manuscripts

3. **Gu, Y.**; Razavi, S.; Habibi, S. Queue-Informed Signal Timing Optimization Under Congested Traffic. Under review at *Transportation Research Part B: Methodological*, 2026.

Co-Authored Journal Articles

4. Sun, K.; **Gu, Y.**; Ma, K.W.F.; Zheng, C.†; Wu, F. Medical Supplies Delivery Route Optimization under Public Health Emergencies Incorporating Metro-based Logistics System. *Transportation Research Record*, 2024. doi:10.1177/03611981231223287.
5. Zheng, C.†; Sun, K.; **Gu, Y.**; Shen, J.; Du, M. Multimodal Transport Path Selection of Cold Chain Logistics Based on Improved Particle Swarm Optimization Algorithm. *Journal of Advanced Transportation*, 2022. doi:10.1155/2022/5458760.
6. Zheng, C.†; **Gu, Y.**; Shen, J.; Du, M. Urban Logistics Delivery Route Planning Based on a Single Metro Line. *IEEE Access*, 2021. doi:10.1109/ACCESS.2021.3069415.
7. Zheng, C.†; **Gu, Y.**; Shen, J.; Du, M.; Ma, G.; Sun, K. Comprehensive Evaluation of Urban Integrated Transportation Network with Multivariate Statistical Analysis: A Case Study of Haimen. *IEEE Access*, 2021. doi:10.1109/ACCESS.2021.3132169.

Conference Papers and Presentations

- **Gu, Y.**; Razavi, S.; Habibi, S. Cycle Queue Length Estimation at Signalized Intersections. CTRF Conference, 2024.
- Presentations: Green Intelligent Transportation System and Safety, 2020 and 2021.

†Master's thesis supervisor.

PATENTS AND SOFTWARE INVENTION

- Patent: A Prediction Method of Short-term Remaining Parking Spaces Based on CNN, CN110503104B.
- Patent: A New Type of Smart Expressway Optoelectronic Delineator, CN208701530U.
- Software copyright: Taxi Operation Analysis System Based on MariaDB V1.0, Registration No. 2019SR0559116.

HONORS, AWARDS, AND LEADERSHIP

Awards and Distinctions

- Excellent Master's Thesis, Hohai University, 2023.
- First-class Scholarship, Excellent Post-graduate Student, and Distinguished Graduate Cadre, Hohai University, multiple awards.
- National Scholarship, Ministry of Education of China, 2017; China Aerospace Science and Technology Corporation Public Welfare Scholarship, 2018.
- Excellent Graduate, Merit Student, First-class Scholarship, and Distinguished Student Cadre, Guilin University of Electronic Technology.

Selected Student Research Competitions

- Challenge Cup National College Students' Extracurricular Academic Science and Technology Contest, Shanghai, China: national third prize and provincial first prize for shared-transportation policy research, 2017.
- Transport Science and Technology Competition, Changsha, China: first prizes for an urban shared-parking charge management system and smart expressway optoelectronic delineator, 2018.

Leadership and Service

- Graduate Association leadership, Hohai University: Presidium member, Deputy Head of Quality Development Department, and Assistant, 2019–2022.

REFERENCES

Available upon request.